

Understanding Multilateral Tariffs and Exchange Rates

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1 Introduction

The conventional wisdom from a standard two-country model of international trade is that a tariff shifts demand toward domestic goods, improving the tariffing country's trade balance and appreciating its currency under the Marshall–Lerner condition. With more than two countries, however, a tariff on one partner also reallocates expenditure across foreign suppliers, so bilateral exchange rates need not move in the same direction. When differentiated tariffs are imposed on multiple foreign suppliers, these effects interact and become harder to characterize. This paper revisits the conventional two-country result and develops a framework for understanding how tariffs affect bilateral and effective exchange rates in a multilateral setting.

In a general N -country trade environment, we show that the log-linearized exchange-rate response to tariffs can be represented as $(-J)^{-1}F$, where F gives the trade-balance effects of tariffs at unchanged exchange rates and J is the Jacobian of trade balances with respect to exchange rates. This representation separates the initial trade-balance effects of the tariffs from the exchange-rate adjustment required to restore equilibrium. We introduce a multilateral Marshall–Lerner (MML) condition on J to characterize this adjustment and show that, under partner symmetry, the response to any tariff vector decomposes into average protection and relative treatment. Average protection determines the change in the tariffing country's nominal effective exchange rate, while relative treatment determines the distribution of bilateral exchange rates around that effective rate.

To quantify these effects in a parsimonious framework, we develop a multilateral trade model with nested-CES preferences. The key determinants of the exchange-rate responses are two substitution margins: η , the elasticity of substitution between domestic and imported tradables, and ρ , the elasticity of substitution across foreign tradables. For a unilateral tariff, we show that the tariffing country always appreciates against the targeted country, while its exchange rate against an untariffed bystander can depreciate when substitution across foreign suppliers is sufficiently strong relative to domestic-import substitution and the associated expenditure effects. We then consider bilateral retaliation and retaliation by all partners and show that retaliation lowers the threshold for depreciation against bystander countries by adding a relative-treatment effect that offsets the average-protection force toward appreciation. Finally, we relate these results to the preferential-trade-agreement literature on third-country terms-of-trade effects. We show that standard competing-exporter models can be understood as deliberately abstracting from some of the domestic-substitution and income channels present in our model in order to isolate substitution

across foreign suppliers.

2 Tariffs and Exchange-Rate Adjustment

This section develops the general N -country exchange-rate adjustment representation, introduces its multilateral Marshall–Lerner condition, and derives the implications of partner symmetry for a general tariff-change vector. We begin with the bilateral case and then extend the same logic to a general differentiable trade system.

2.1 The bilateral case

Consider first two countries, A and B . Let e_{AB} denote units of A 's currency per unit of B 's, so that a rise in e_{AB} is a depreciation of A . Country A levies a tariff τ on imports from B . Both the tariff and the exchange rate affect relative consumer prices and hence A 's trade balance. The balanced-trade equilibrium is therefore defined by

$$\text{TB}_A(e_{AB}, \tau) = 0.$$

Write $\nu \equiv \log e_{AB}$. Linearizing the trade-balance condition around an equilibrium gives

$$d\text{TB}_A = \text{TB}_{A\nu} d\nu + \text{TB}_{A\tau} d\tau, \quad \text{TB}_{A\nu} \equiv \frac{\partial \text{TB}_A}{\partial \nu}, \quad \text{TB}_{A\tau} \equiv \frac{\partial \text{TB}_A}{\partial \tau} \Big|_{\nu \text{ fixed}}. \quad (1)$$

Suppose that $\text{TB}_{A\tau} > 0$, i.e., the tariff improves A 's trade balance at unchanged exchange rates. This is the standard case in which the tariff raises the domestic price of imports and shifts expenditure away from foreign goods, reducing import demand relative to exports. Restoring balance then requires

$$d\nu^* = -\frac{\text{TB}_{A\tau}}{\text{TB}_{A\nu}} d\tau. \quad (2)$$

For a tariff increase $d\tau > 0$, this is an appreciation if and only if $\text{TB}_{A\nu} > 0$, that is, if and only if a depreciation improves A 's trade balance. At a balanced-trade baseline,

$$\text{TB}_{A\nu} = X(\varepsilon_X + \varepsilon_M - 1),$$

where X is the value of exports, and ε_X and ε_M are the export and import demand elasticities, respectively. Hence a depreciation improves A 's trade balance under the bilateral Marshall–Lerner condition

$$\varepsilon_X + \varepsilon_M > 1. \quad (3)$$

Figure 1 illustrates the logic of Equations (1)–(2). The tariff first changes the trade balance at unchanged exchange rates, and under the bilateral Marshall–Lerner condition (3), the exchange rate then appreciates to restore balance.

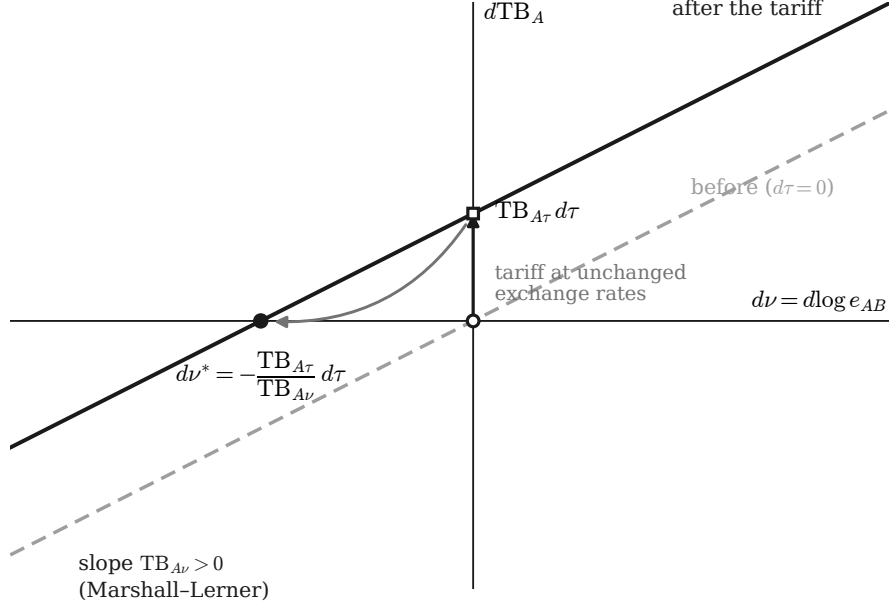


Figure 1: Bilateral tariff and exchange-rate adjustment under $TB_{A\tau} > 0$ and $TB_{A\nu} > 0$.

2.2 The local disturbance–adjustment representation

Now consider a differentiable N -country trade environment. Taking A 's currency as the numeraire, the $N - 1$ bilateral exchange rates e_{Aj} , $j \neq A$, span all exchange-rate changes, and Walras' law leaves $N - 1$ independent trade-balance conditions. We therefore characterize the balanced-trade equilibrium by

$$TB_i(\{e_{Aj}\}_{j \neq A}, \{\tau_{jk}\}_{j \neq k}) = 0, \quad i \neq A.$$

Stack the log exchange rates as $\boldsymbol{\nu} \equiv (\nu_j)_{j \neq A}$, with $\nu_j \equiv \log e_{Aj}$, and the bilateral tariff changes as $d\boldsymbol{\tau} \equiv (d\tau_{jk})_{j \neq k}$. Linearizing the $N - 1$ trade-balance conditions gives

$$J d\boldsymbol{\nu} + G d\boldsymbol{\tau} = 0, \quad J_{il} \equiv \frac{\partial TB_i}{\partial \nu_l}, \quad G_{i,(jk)} \equiv \frac{\partial TB_i}{\partial \tau_{jk}} \Big|_{\boldsymbol{\nu} \text{ fixed}}. \quad (4)$$

Now separate a general tariff change into its direction and scale. Let $\mathbf{a}$ denote the direction of the tariff change, normalized so that $\|\mathbf{a}\|_2 = 1$, and let $ds \geq 0$ denote its scale, so that

$$d\boldsymbol{\tau} = \mathbf{a} ds.$$

The vector $\mathbf{a}$ therefore describes which bilateral tariffs change and in what proportions, while $ds = \|d\boldsymbol{\tau}\|_2$ gives the magnitude of the tariff change. For a given direction $\mathbf{a}$, define

$$\mathbf{F} \equiv G \mathbf{a}, \quad (5)$$

where $\mathbf{F}$ is the vector of trade-balance disturbances generated by a unit increase in the scale of the

tariff change at unchanged exchange rates. Substituting $d\tau = \mathbf{a} ds$ into (4) gives

$$J d\boldsymbol{\nu} + \mathbf{F} ds = 0. \quad (6)$$

Hence the marginal exchange-rate response along tariff-change direction $\mathbf{a}$ is

$$\frac{d\boldsymbol{\nu}}{ds} = (-J)^{-1} \mathbf{F}. \quad (7)$$

Equation (7) is the multilateral analogue of the bilateral adjustment in Equation (2) (setting $N = 2$, $\mathbf{a} = 1$, and $ds = d\tau$). This expression shows that the exchange-rate response to tariffs can be separated into two parts. The vector $\mathbf{F}$, and more generally G , describes the initial trade-balance effects of tariff changes at unchanged exchange rates, including substitution and income effects. The matrix J describes how trade balances respond to exchange-rate changes, while $(-J)^{-1}$ maps the initial disturbance into the exchange-rate adjustment required to restore equilibrium.

2.3 The multilateral Marshall–Lerner condition

In the bilateral case, the requirement that exchange-rate adjustment stabilizes the trade balance is the scalar condition $-J = \text{TB}_{A\nu} > 0$, which has a straightforward interpretation. In the multilateral case, a tariff generates a vector of trade-balance disturbances, and the exchange-rate response depends on the full adjustment mapping $(-J)^{-1}$. A natural analogue of the bilateral Marshall–Lerner condition is therefore that exchange-rate cross-effects have the stabilizing sign and that the resulting adjustment mapping be well defined and preserve the sign of trade-balance disturbances.

Definition 2.1 (Multilateral Marshall–Lerner condition). *The adjustment mechanism satisfies the multilateral Marshall–Lerner condition (MML) if*

$$J_{il} \geq 0 \quad (l \neq i), \quad (-J)^{-1} \text{ exists and satisfies } (-J)^{-1} \geq 0. \quad (8)$$

Equivalently, $-J$ is a nonsingular M-matrix.

Under MML, each positive component of the trade-balance disturbance makes a weakly positive contribution to every bilateral exchange-rate response. By (7), writing $\vartheta_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0$,

$$\frac{d\nu_i}{ds} = \sum_j \vartheta_{ij} F_j, \quad (9)$$

where ϑ_{ij} measures the response of the A - i bilateral exchange rate to a unit trade-balance disturbance in country j , with a positive response denoting a depreciation of A against i . A uniformly signed $\mathbf{F}$ therefore cannot generate exchange-rate responses of the opposite sign. When $\mathbf{F}$ has mixed signs, the response of any individual exchange rate depends on the full vector of trade-balance effects rather than on the corresponding element F_i alone.

The MML condition stated directly in terms of $(-J)^{-1}$ is somewhat abstract. A more intuitive sufficient condition can be stated in terms of gross substitutability. To express this condition

without choosing a numeraire, extend the notation and let ν_i , $i = 1, \dots, N$, denote the log value of currency i in a common unit of account, so that

$$d \log e_{ij} = d\nu_j - d\nu_i,$$

and let $\tilde{J}_{il} \equiv \partial \text{TB}_i / \partial \nu_l$ be the full $N \times N$ Jacobian. Homogeneity of each trade balance in currency values implies, by Euler's theorem, that

$$\sum_l \tilde{J}_{il} = \text{TB}_i,$$

so at a balanced equilibrium

$$\sum_l \tilde{J}_{il} = 0 \quad \forall i.$$

Separately, $\sum_i \text{TB}_i \equiv 0$ by Walras' law implies

$$\sum_i \tilde{J}_{il} = 0 \quad \forall l.$$

Thus,

$$\sum_l \tilde{J}_{il} = 0 \quad \forall i, \quad \sum_i \tilde{J}_{il} = 0 \quad \forall l. \quad (10)$$

The row identity pins down each diagonal entry of $\tilde{J}$ from the off-diagonal entries, so the relevant sign restriction can be stated entirely in terms of these cross-effects.

Assumption 1 (Gross substitutability of trade balances). $\tilde{J}_{il} \geq 0$ for all $l \neq i$.

An appreciation of currency l raises the price of l 's goods relative to those produced elsewhere. Assumption 1 requires that this weakly improve every other country's trade balance once the associated income and expenditure effects are included. The intuition is the same as in the bilateral Marshall–Lerner condition: exchange-rate movements operate in the direction needed to offset trade-balance disturbances. Together with the row identity in (10), this sign structure makes the reduced matrix weakly diagonally dominant; the additional connectivity conditions in Lemma 2.2 ensure that $-J$ is a nonsingular M-matrix.

Lemma 2.2 (Sufficiency of gross substitutability). *Suppose each trade balance is homogeneous of degree one in currency values. Under Assumption 1, with J irreducible and $\tilde{J}_{iA} > 0$ for some i , MML (8) holds.*

Thus, under gross substitutability and the conditions of Lemma 2.2, the multilateral Marshall–Lerner condition follows from economically interpretable cross-effects of exchange rates on trade balances. The proof is given in Appendix F.

2.4 Symmetric partners

Now restrict attention to tariff changes imposed by A , and write $\mathbf{t} = (t_j)_{j \neq A}$, with $t_j \equiv d\tau_{Aj}$. Restricting G to these $N - 1$ tariff changes, let G henceforth denote the resulting $(N - 1) \times (N - 1)$

matrix. Suppose that A 's trading partners are symmetric, so that relabeling them leaves the reduced system unchanged:

$$PJP' = J, \quad PGP' = G$$

for every permutation matrix P of the $N - 1$ partners.

Under partner symmetry, all partners are interchangeable except for how they are treated by A 's tariff changes. It is therefore natural to separate the part of the tariff change that is common across partners from the part that distinguishes among them. We refer to these as *average protection* and *relative treatment*, respectively. Decompose

$$\mathbf{t} = \bar{t} \mathbf{1} + \tilde{\mathbf{t}}, \quad \bar{t} \equiv \frac{1}{N-1} \mathbf{1}' \mathbf{t}, \quad \mathbf{1}' \tilde{\mathbf{t}} = 0. \quad (11)$$

Componentwise,

$$t_j = \bar{t} + (t_j - \bar{t}), \quad \tilde{t}_j \equiv t_j - \bar{t}. \quad (12)$$

Thus $\bar{t} \mathbf{1}$ is the common component of the tariff change across partners, while $\tilde{\mathbf{t}}$ records each partner's treatment relative to that common component.

Now define

$$d\nu_A^E \equiv \frac{1}{N-1} \sum_{j \neq A} d\nu_j,$$

which denotes the equally weighted average of A 's bilateral exchange-rate responses. At a symmetric baseline with equal bilateral trade weights, this coincides with the change in A 's nominal effective exchange rate in (21). The following result shows that partner symmetry allows the exchange-rate response to be characterized by the same common and relative components of A 's tariff changes.

Proposition 2.3 (Average protection and relative treatment). *Suppose A 's partners are symmetric and $N \geq 3$. Then $-J$ and G act separately on the common subspace $\text{span}\{\mathbf{1}\}$ and the zero-sum subspace $\{v : \mathbf{1}'v = 0\}$. Let j_S, j_D and g_S, g_D denote their respective scalars on these two subspaces. If $j_S, j_D \neq 0$, the response to any tariff-change vector $\mathbf{t}$ is*

$$d\nu = \lambda_S \bar{t} \mathbf{1} + \lambda_D \tilde{\mathbf{t}}, \quad \lambda_S \equiv \frac{g_S}{j_S}, \quad \lambda_D \equiv \frac{g_D}{j_D}. \quad (13)$$

Hence

$$d\nu_A^E = \lambda_S \bar{t}, \quad d\nu_j - d\nu_A^E = \lambda_D \tilde{t}_j, \quad d \log e_{jk} = \lambda_D (t_k - t_j), \quad (14)$$

so average protection determines the common exchange-rate response, while relative treatment determines bilateral deviations and cross rates. Furthermore, MML (8) is equivalent to

$$j_S > 0, \quad j_D \geq j_S. \quad (15)$$

The two-term response follows directly from partner symmetry. Any matrix invariant to relabeling the partners has the form $aI + b\mathbf{1}\mathbf{1}'$. Since $\mathbf{1}'\mathbf{1} = N - 1$ and $\mathbf{1}'\tilde{\mathbf{t}} = 0$,

$$(aI + b\mathbf{1}\mathbf{1}') \mathbf{t} = (a + (N-1)b) \bar{t} \mathbf{1} + a \tilde{\mathbf{t}}.$$

Thus the matrix acts separately on the common and zero-sum components, with eigenvalues $a + (N - 1)b$ and a , respectively. For $-J$ and G , these are

$$\begin{aligned} -J &= a_J I + b_J \mathbf{1}\mathbf{1}' : & j_S &= a_J + (N - 1)b_J, & j_D &= a_J, \\ G &= a_G I + b_G \mathbf{1}\mathbf{1}' : & g_S &= a_G + (N - 1)b_G, & g_D &= a_G. \end{aligned} \tag{16}$$

Because relative treatment sums to zero, it has no effect on the equally weighted average of the bilateral exchange-rate responses, giving the first equality in (14). The MML characterization in (15) follows from the same structure. Since

$$b_J = \frac{j_S - j_D}{N - 1},$$

$j_D \geq j_S$ is equivalent to $b_J \leq 0$, which gives the required nonnegative off-diagonal elements of J . Together with $j_S > 0$, this also implies $(-J)^{-1} \geq 0$. The formal proof is given in Appendix F.

3 The Nested-CES Model

We now specify an N -country nested-CES model that gives economic structure to the general results of Section 2.

3.1 Production, preferences, and equilibrium

We index countries by $i, j \in \{1, \dots, N\}$. Where appropriate, we also use A for the tariffing country, B for the tariff target, and C for an untariffed bystander. Each country produces a country-specific tradable good and a non-tradable good using labor only, with linear production functions

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i.$$

Normalizing the producer price of country i 's tradable to one in i 's currency, perfect competition then implies

$$w_i = A_{T_i}, \quad P_{N_i} = A_{T_i} / A_{N_i}.$$

Linear production and the producer-price normalization therefore fix wages and domestic producer prices in own currency, so relative international producer prices adjust solely through exchange rates.

We consider nested-CES preferences that allow substitution between domestic and imported tradables to differ from substitution across foreign suppliers:

$$\begin{aligned} C_i &= C_{T_i}^{\alpha_T} C_{N_i}^{1-\alpha_T} && \text{(tradable share } \alpha_T), \\ C_{T_i} &= \left[\alpha_D^{1/\eta} C_{T_i i}^{\frac{\eta-1}{\eta}} + (1 - \alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} && \text{(home weight } \alpha_D, \text{ elasticity } \eta), \\ M_i &= \left[\sum_{j \neq i} \beta_{ij}^{1/\rho} C_{T_j i}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} && \text{(origin weights } \beta_{ij}, \text{ elasticity } \rho). \end{aligned}$$

The top layer of the nested utility function is Cobb-Douglas, so the expenditure share on tradables is fixed at α_T and the share on non-tradables at $1 - \alpha_T$. Within tradables, η governs substitution between the domestic tradable and the import composite, while ρ governs substitution across foreign origins. The latter margin is inactive at $N = 2$, when there is only one foreign supplier, and becomes relevant at $N \geq 3$.

The *symmetric benchmark* imposes $\beta_{ij} = 1/(N - 1)$, common parameters $(\alpha_T, \alpha_D, \eta, \rho)$, and equal endowments and productivities. Symmetry across countries does not eliminate home bias: domestic tradables retain weight α_D . Full symmetry between the domestic tradable and each foreign tradable would additionally require $\rho = \eta$ and $\alpha_D = 1/N$, which we do not impose.

As in Section 2, let e_{ij} denote units of i 's currency per unit of j 's, and take A 's currency as the numeraire, with $\nu_j \equiv \log e_{Aj}$ and $d\nu_j = d \log e_{Aj}$. We allow for general ad valorem tariffs $\tau_{ij} \geq 0$ levied by country i on imports from country j , with $\tau_{ii} = 0$. Since the producer price of country j 's tradable is normalized to one in j 's currency, its consumer price in country i is $p_{ij} = (1 + \tau_{ij})e_{ij}$, while $p_{ii} = 1$. The corresponding import, tradable, and aggregate price indices are

$$P_{M_i} = \left[\sum_{j \neq i} \beta_{ij} p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \quad P_{T_i} = \left[\alpha_D p_{ii}^{1-\eta} + (1 - \alpha_D) P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad P_i = \kappa_i P_{T_i}^{\alpha_T} P_{N_i}^{1-\alpha_T}. \quad (17)$$

The associated tradable expenditure shares of income are

$$s_{ii} = \alpha_T \alpha_D \left(\frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \quad s_{ij} = \alpha_T (1 - \alpha_D) \left(\frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \quad j \neq i. \quad (18)$$

Thus $s_{ij} I_i$ is tariff-inclusive expenditure on imports from j , while $s_{ij} I_i / (1 + \tau_{ij})$ is the corresponding tariff-exclusive border value.

Household income consists of labor income, $w_i L_i$, and tariff revenue, which is rebated lump-sum

$$I_i = \frac{w_i L_i}{1 - \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij}}. \quad (19)$$

At the free-trade baseline, $I_i = w_i L_i$, so the Cobb-Douglas top layer implies that equilibrium labor allocations are α_T and $1 - \alpha_T$ in the tradable and non-tradable sectors, respectively, independently of other production or preference parameters and exchange rates. With tariffs, rebated tariff revenue raises income relative to labor income, so these labor allocations are no longer fixed.

To express bilateral flows in a common currency, define $\tilde{I}_i \equiv e_{Ai} I_i$. Since tariff revenue is a domestic transfer, trade balances are measured at tariff-exclusive border prices:

$$\text{TB}_i = \sum_{k \neq i} f_{ki} - \sum_{j \neq i} f_{ij}, \quad f_{ij} \equiv \frac{s_{ij} \tilde{I}_i}{1 + \tau_{ij}}, \quad (20)$$

where f_{ij} is the border value, in A 's currency, of country i 's imports from country j . As in Section 2, Walras' law leaves $N - 1$ independent trade-balance conditions. With no internationally traded assets, equilibrium requires

$$\text{TB}_i(\{e_{Aj}\}_{j \neq A}, \{\tau_{jk}\}_{j \neq k}) = 0, \quad i \neq A,$$

which, for a given set of tariffs, determine the $N - 1$ equilibrium exchange rates relative to A .

Define country i 's nominal effective exchange rate using baseline bilateral trade weights as

$$E_i \equiv \prod_{j \neq i} e_{ij}^{w_{ij}}, \quad w_{ij} = \frac{f_{ij} + f_{ji}}{\sum_{k \neq i} (f_{ik} + f_{ki})}, \quad (21)$$

where the weights are evaluated at the baseline equilibrium. At the symmetric benchmark, $w_{ij} = 1/(N - 1)$, so the log nominal effective exchange rate is the simple average of country i 's bilateral log exchange rates.

Finally, define bilateral terms of trade and real exchange rates as

$$\text{ToT}_{ij} \equiv \frac{1}{e_{ij}}, \quad q_{ij} \equiv \frac{e_{ij} P_j}{P_i}. \quad (22)$$

An appreciation of country i against country j therefore corresponds to an improvement in i 's bilateral terms of trade.

3.2 The two-country case

Appendix B gives the general N -country linearization, including the $N = 2$ calculations used here. Write $m \equiv \alpha_T(1 - \alpha_D)$ for the import share of income, and suppose A raises its tariff on B by $d\tau$. With only one foreign supplier, ρ is inactive. Define

$$\rho_1^* \equiv \alpha_D \eta + (1 - \alpha_D)(1 - \alpha_T) > 0, \quad (23)$$

and

$$D_2 \equiv 2\alpha_D(\eta - 1) + 1. \quad (24)$$

Linearizing (20) at the balanced free-trade baseline gives

$$d\text{TB}_A = mD_2 d\nu_B + m\rho_1^* d\tau = 0, \quad (25)$$

and therefore

$$\left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0} = -\frac{\rho_1^*}{D_2}. \quad (26)$$

In the terminology of Section 2.1, ρ_1^* and D_2 determine, up to the common scale factor m , the tariff-induced trade-balance disturbance and the exchange-rate adjustment, respectively. Here $\text{TB}_{A\tau} = m\rho_1^* > 0$, so the tariff unambiguously improves A 's trade balance at unchanged exchange rates. The expression for ρ_1^* combines substitution between the domestic tradable and imports, measured by $\alpha_D \eta$, with the associated income and expenditure effect falling on nontradables, measured by $(1 - \alpha_D)(1 - \alpha_T)$.

At the symmetric two-country benchmark, $\varepsilon_X = \varepsilon_M = 1 + \alpha_D(\eta - 1)$, so $D_2 = \varepsilon_X + \varepsilon_M - 1$. Hence the Marshall–Lerner condition of Section 2.1 is $D_2 > 0$, which holds whenever $\eta \geq 1$. Appendix A gives the exact finite-tariff Cobb–Douglas solution.

3.3 The N -country case

Now consider the N -country case, where A 's trading partners are symmetric in the sense of Section 2.4. Proposition 2.3 shows that the exchange-rate response is characterized by its average-protection and relative-treatment components. We now derive closed-form expressions for the corresponding coefficients under the nested-CES structure.

Theorem 3.1 (Exchange-rate responses in the symmetric nested-CES model). *Consider the nested-CES model of Section 3 at the symmetric benchmark. Let $\mathbf{t} = (t_j)_{j \neq A}$ denote any vector of tariff changes imposed by A , with average $\bar{t}$ as in (12). Define*

$$D_N \equiv N\alpha_D(\eta - 1) + (N - 2)\rho + 1. \quad (27)$$

The MML condition (8) holds if and only if $D_N > 0$. Since $\alpha_D > 0$ and $\rho > 0$, $\eta \geq 1$ is sufficient for $D_N > 0$.

When $D_N > 0$, the average-protection and relative-treatment coefficients are

$$\lambda_S = -\frac{(N - 1)\rho_1^*}{D_N}, \quad \lambda_D = -\frac{(N - 1)\rho}{ND_N},$$

and the equilibrium exchange-rate response is

$$d\nu_j = \lambda_S \bar{t} + \lambda_D \tilde{t}_j. \quad (28)$$

For $N = 2$, there is only one trading partner, so relative treatment is mechanically zero, $D_N = D_2$, and Theorem 3.1 reduces to the two-country result in Section 3.2. For $N \geq 3$, relative treatment is governed by $\lambda_D < 0$, implying that A appreciates more against partners facing larger tariff increases. The proof is given in Appendix B.

Observe also that average protection and relative treatment enter differently through the tariff-induced trade-balance disturbance and the exchange-rate adjustment. For the trade-balance disturbance,

$$g_S = -\frac{m\rho_1^*}{N - 1}, \quad g_D = -\frac{m\rho}{N - 1}, \quad (29)$$

while for the trade-balance response to exchange-rate movements,

$$j_S = \frac{mD_N}{(N - 1)^2}, \quad j_D = N \frac{mD_N}{(N - 1)^2}. \quad (30)$$

Comparing the two components gives

$$\frac{g_D}{g_S} = \frac{\rho}{\rho_1^*}, \quad \frac{j_D}{j_S} = N. \quad (31)$$

The two ratios summarize how average protection and relative treatment differ at the disturbance and adjustment stages. The first ratio in (31) compares their effects on trade balances at unchanged exchange rates. Both scale with the bilateral import share $m/(N - 1)$, so N drops out of the ratio. Average protection operates through substitution between domestic and imported

goods and the associated expenditure effects, summarized by ρ_1^* , while relative treatment operates through substitution across foreign suppliers and is governed by ρ .

The distinction between average protection and relative treatment also carries over to domestic allocation. At the symmetric free-trade baseline, a pure relative-treatment change leaves average protection, tariff revenue, and income unchanged to first order, so it does not affect the allocation of labor between tradables and non-tradables. By contrast, an increase in average protection raises tariff revenue and income, shifting labor toward non-tradables.

The second ratio in (31) compares the responsiveness of trade balances to exchange-rate movements in the two directions. A common exchange-rate movement changes A 's currency against all partners together, while a relative movement also changes competitiveness across foreign suppliers. Trade balances are therefore N times more responsive in the relative-treatment direction, so a disturbance of a given size requires only $1/N$ as much exchange-rate adjustment.

Combining the two ratios gives

$$\frac{\lambda_D}{\lambda_S} = \frac{g_D}{g_S} \frac{j_S}{j_D} = \frac{\rho}{N\rho_1^*}. \quad (32)$$

Thus N does not affect the relative size of the initial disturbances, but it reduces the relative-treatment exchange-rate response relative to the average-protection response.

3.4 A unilateral tariff

Suppose now that A raises its tariff on B alone, so that $\mathbf{t} = d\tau \mathbf{e}_B$. Then

$$\bar{t} = \frac{d\tau}{N-1}, \quad \tilde{t}_B = \frac{N-2}{N-1}d\tau, \quad \tilde{t}_C = -\frac{d\tau}{N-1} \quad \text{for each bystander } C.$$

Since only average protection affects the effective exchange rate,

$$\frac{d\nu_A^E}{d\tau} = -\frac{\rho_1^*}{D_N} < 0, \quad (33)$$

so a unilateral tariff always appreciates A 's effective exchange rate at the symmetric benchmark, for every $N \geq 2$.

The bilateral responses additionally depend on relative treatment. Against the targeted country, average protection and relative treatment reinforce each other:

$$\frac{d \log e_{AB}}{d\tau} = -\frac{N\rho_1^* + (N-2)\rho}{ND_N} < 0, \quad (34)$$

so A always appreciates against B , again for every $N \geq 2$. For $N \geq 3$ there is also an untargeted bystander C , for which the two components work in opposite directions:

$$\frac{d \log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{ND_N}. \quad (35)$$

Average protection pushes A to appreciate against C , while C 's favorable relative treatment pushes

in the opposite direction. Hence A depreciates against C if and only if

$$\rho > N\rho_1^*. \quad (36)$$

The threshold highlights the distinct roles of the two substitution elasticities. A larger η strengthens substitution from imports toward the domestic tradable and raises ρ_1^* , making a bystander depreciation less likely, while a larger ρ strengthens substitution across foreign suppliers and makes it more likely. The factor N reflects the adjustment ratio derived in Section 3.3. Thus condition (36) requires the relative-treatment effect to be strong enough to offset the average-protection response after accounting for the greater responsiveness of trade balances in the relative-treatment direction.

In particular, the reversal cannot occur when the two substitution elasticities coincide and $\alpha_D \geq 1/N$. If $\rho = \eta = \sigma$, then

$$N\rho_1^* - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T), \quad (37)$$

which is nonnegative for $\alpha_D \geq 1/N$. Hence $\rho \leq N\rho_1^*$ and the reversal condition cannot hold.¹ More generally, for $\alpha_D \geq 1/N$,

$$N\rho_1^* - \eta = \eta(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) \geq 0,$$

so $\rho > \eta$ is necessary, but not sufficient, for a bystander depreciation. Appendix C gives the corresponding real exchange-rate results.

Note that A may still appreciate against a bystander even if the bystander's trade balance improves at unchanged exchange rates. From (51),

$$F_C = \frac{m(\rho - \rho_1^*)}{(N - 1)^2}, \quad (38)$$

so

$$F_C > 0 \iff \rho > \rho_1^*, \quad \text{whereas} \quad d\nu_C > 0 \iff \rho > N\rho_1^*.$$

The first threshold depends only on the relative size of the tariff disturbances, g_D/g_S , while the second also reflects the adjustment ratio $j_D/j_S = N$. Thus an improvement in the bystander's trade balance at unchanged exchange rates need not be large enough to reverse its bilateral exchange-rate response. Figure 2 illustrates the distinction.

¹Equality in (37) requires $\alpha_D = 1/N$ and $\alpha_T = 1$. At $\alpha_D = 1/N$ with $\rho = \eta = \sigma$, $d \log e_{AC}/d\tau = -(1 - \alpha_T)/(N\sigma)$, which approaches zero from below as $\alpha_T \rightarrow 1$; see Appendix B.

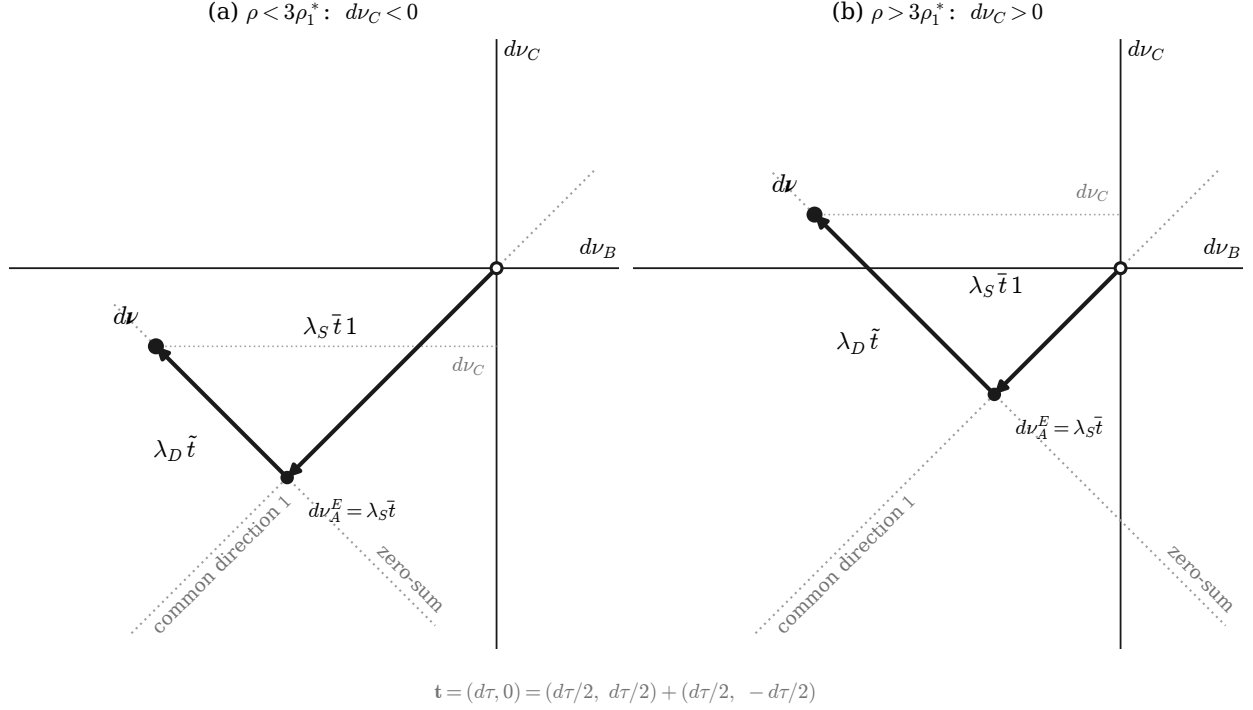


Figure 2: Average protection and relative treatment under a unilateral tariff at $N = 3$. Panel (a): $\rho < 3\rho_1^*$; panel (b): $\rho > 3\rho_1^*$.

4 Trade Wars and Preferential Tariffs

Because our analysis is local, simultaneous tariff changes by several countries can be superposed to first order. At the symmetric benchmark, each tariffing country's contribution follows from Theorem 3.1.

4.1 Trade wars

We first consider symmetric bilateral retaliation. Suppose A raises its tariff on B by $d\tau$ and B raises its tariff on A by the same amount. By symmetry, the two effects on the A – B bilateral rate cancel. For any bystander C , superposing the two unilateral responses gives

$$\frac{d \log e_{AB}}{d\tau} = 0, \quad \frac{d \log e_{AC}}{d\tau} = \frac{\rho - \rho_1^*}{D_N}, \quad \frac{d\nu_A^E}{d\tau} = \frac{N-2}{N-1} \frac{\rho - \rho_1^*}{D_N}. \quad (39)$$

Compared with a unilateral tariff, retaliation lowers the bystander threshold from $N\rho_1^*$ to ρ_1^* , exactly the threshold at which the bystander's trade balance improves at unchanged exchange rates. With symmetric retaliation, B 's tariff on A adds the relative-treatment contribution needed to offset the adjustment-ratio penalty that generates the factor N under a unilateral tariff. The sign of the bystander's exchange-rate response therefore follows the sign of its trade-balance disturbance at unchanged exchange rates.

Now suppose that A raises its tariff by $d\tau$ on every partner and every partner raises its tariff

by $d\tau$ on A . This configuration is the sum of $N - 1$ bilateral trade wars involving A . For any A - j bilateral rate, the bilateral war between A and j contributes zero by (39), while each of the remaining $N - 2$ wars contributes the same bystander response.

Proposition 4.1 (Trade wars). *At the symmetric benchmark of the nested-CES model, with $N \geq 3$ and under MML:*

- (i) *Under symmetric bilateral retaliation between A and B , the belligerents' effective exchange rates depreciate, and every bystander appreciates against both, if and only if $\rho > \rho_1^*$.*
- (ii) *If A tariffs every partner and every partner tariffs A by the same amount, then*

$$\frac{d \log e_{Aj}}{d\tau} = \frac{d\nu_A^E}{d\tau} = \frac{(N-2)(\rho - \rho_1^*)}{D_N} \quad \forall j, \quad (40)$$

while all exchange rates among A 's partners are unchanged. Hence A depreciates against every partner if and only if $\rho > \rho_1^$.*

Thus the same threshold applies under bilateral retaliation and retaliation by all partners.

4.2 Preferential tariffs and competing exporters

A standard benchmark in the preferential-trade-agreement literature has one importing country and two competing exporters, following Viner (1950) and Mundell (1964). Importer A is endowed only with the numeraire good y , while exporters B and C are endowed with good x . Preferences are quasilinear, $U_i = u_i(x_i) + y_i$. Country A levies specific tariffs t_B and t_C . Let p be the domestic price of x in A , and let q_j be exporter j 's price of x in terms of the numeraire, which is j 's terms of trade. Market clearing requires

$$p = q_B + t_B = q_C + t_C, \quad M_A(p) = E_B(p - t_B) + E_C(p - t_C), \quad M'_A < 0, \quad E'_j > 0, \quad (41)$$

where $M_A(p) = x_A^d(p)$ and $E_j(q_j) = \omega_j - x_j^d(q_j)$. Differentiating with respect to t_B gives

$$\frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0, 1), \quad \frac{dq_B}{dt_B} = -\frac{E'_C - M'_A}{E'_B + E'_C - M'_A} < 0, \quad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \quad (42)$$

Thus a tariff on imports from B worsens B 's terms of trade and improves C 's.

In our model, $\text{ToT}_{CA} = e_{AC}$ by (22), so an improvement in C 's terms of trade corresponds to an increase in e_{AC} . In the competing-exporter model this occurs unconditionally after a tariff on B , whereas at the symmetric free-trade benchmark of the nested-CES model (36) requires $\rho > N\rho_1^*$. A differential tariff change in the opposite direction reverses these signs locally: for $\mathbf{t} = -d\tau \mathbf{e}_B$, a non-member C 's terms of trade worsen if and only if $\rho > N\rho_1^*$. This captures the mechanism behind preferential liberalization, although the closed-form threshold is derived around free trade rather than around a positive common external tariff.

The difference reflects the economic environment. In the competing-exporter model, A does not produce the importable good, so a tariff reallocates demand only across the two foreign suppliers.

In our model, substitution across foreign origins, governed by ρ , operates alongside substitution toward the domestic tradable and the associated income effects, summarized by ρ_1^* . The competing-exporter sign therefore obtains only when substitution across foreign suppliers is sufficiently strong relative to these domestic channels.

The two environments are not formally nested. The competing-exporter model has a common importable supplied by B and C and quasilinear preferences, while our model has country-specific differentiated tradables and nested-CES preferences, so the comparison is one of mechanisms rather than special cases. Related competing-exporter models include Bagwell and Staiger (1999) and Ornelas (2005), while Krugman (1991) and Bond and Syropoulos (1996) study related bloc and endowment environments.

5 Conclusion

A tariff affects exchange rates through two steps. At unchanged exchange rates it changes countries' trade balances, summarized by F ; exchange rates then adjust through $(-J)^{-1}$. In the two-country case this reduces to the familiar Marshall–Lerner result: when a tariff improves the trade balance at unchanged exchange rates, the elasticity condition ensures that the tariffing country's currency appreciates.

With three or more countries, a tariff can also redirect expenditure across foreign suppliers. Under symmetry, the response to any tariff vector separates into an average-protection component and a relative-treatment component. Average protection determines the effective exchange-rate response, while relative treatment determines deviations of bilateral rates from that common response. At the symmetric benchmark of the nested-CES model, the tariffing country continues to appreciate against the targeted partner, but its exchange rate against an untariffed partner can move in either direction.

At the symmetric benchmark, a unilateral tariff always appreciates the tariffing country's effective exchange rate, while the bilateral rate against an untariffed bystander reverses only if

$$\rho > N\rho_1^*.$$

The factor N reflects multilateral exchange-rate adjustment rather than the relative size of the initial tariff disturbances. Under symmetric bilateral retaliation, the threshold falls to ρ_1^* and becomes independent of N . Under asymmetric trade weights, even the effective exchange-rate result need not hold (Appendix D).

In the standard competing-exporter benchmark, by contrast, substitution across foreign suppliers generates an unambiguous third-country terms-of-trade effect because the domestic-substitution and associated income channels are absent.

For a unilateral tariff, the familiar two-country appreciation result survives multilateralization most cleanly at the level of the effective exchange rate. Bilateral exchange rates also reflect how trade policy differs across partners, and their responses need not inherit the sign of the aggregate adjustment.

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A Exact Cobb–Douglas solution ($N = 2$)

At $\eta = 1$, expenditure shares are fixed at their weights. From (19), $I_A = w_A L_A / [1 - \frac{\tau}{1+\tau} m]$, and $TB_A = 0$ becomes $e m w_B L_B = m I_A / (1 + \tau)$:

$$e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1 - m)\tau}, \quad \left. \frac{d \log e}{d \tau} \right|_{\tau=0} = -(1 - m) = -\left. \frac{\rho_1^*}{D_2} \right|_{\eta=1}. \quad (43)$$

Thus $e(\tau)$ is decreasing in τ , so the sign of the linearized response in (26) is exact for all tariff levels in this case. Here $\eta = 1$ fixes the elasticity, not the weight α_D .

B The Linearized System

At the balanced free-trade baseline, let $m \equiv \alpha_T(1 - \alpha_D)$, $b_{ij} \equiv \beta_{ij}$, and $f_{ij} = s_{ij} \tilde{I}_i$, with $\sum_k f_{ki} = \sum_j f_{ij}$. All incomes and flows are measured in the common currency. Linearizing the demand, income, and trade-balance equations gives

$$dp_{ij} = d\nu_j - d\nu_i + d\tau_{ij} \quad (dp_{ii} = 0), \quad (44)$$

$$d \log P_{M_i} = \sum_{j \neq i} b_{ij} dp_{ij}, \quad d \log P_{T_i} = (1 - \alpha_D) d \log P_{M_i}, \quad (45)$$

$$d \log s_{ii} = -(1 - \eta)(1 - \alpha_D) d \log P_{M_i}, \quad (46)$$

$$d \log s_{ij} = (1 - \rho)(dp_{ij} - d \log P_{M_i}) + (1 - \eta)\alpha_D d \log P_{M_i}, \quad (47)$$

$$d \log \tilde{I}_i = d\nu_i + \sum_{j \neq i} s_{ij} d\tau_{ij}, \quad (48)$$

$$d TB_i = \sum_{k \neq i} f_{ki} (d \log s_{ki} + d \log \tilde{I}_k - d\tau_{ki}) - \sum_{j \neq i} f_{ij} (d \log s_{ij} + d \log \tilde{I}_i - d\tau_{ij}). \quad (49)$$

These equations determine the model-specific matrices J and G in (4). At $N = 2$ they reduce to the calculation in Section 3.2.

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} D_N \quad (l \neq i), \quad D_N \equiv N\alpha_D(\eta - 1) + (N-2)\rho + 1. \quad (50)$$

Lemma B.1 below gives the derivation.

Lemma B.1 (Full Jacobian at the symmetric baseline). *At the symmetric balanced baseline of Appendix B, where $b_{ij} = 1/(N-1)$ and $f_{ij} = m/(N-1)$, the full $N \times N$ Jacobian of the nested-CES model is*

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} D_N \quad (l \neq i), \quad \tilde{J}_{ii} = -\frac{m}{N-1} D_N, \quad D_N = N\alpha_D(\eta - 1) + (N-2)\rho + 1.$$

Proof. Set $d\tau_{jk} = 0$, so $dp_{ij} = d\nu_j - d\nu_i$ and $d\log \tilde{I}_i = d\nu_i$. Writing $S \equiv \sum_j d\nu_j$, (45) gives

$$d\log P_{M_k} = \frac{S - N d\nu_k}{N-1}, \quad \frac{\partial \log P_{M_k}}{\partial \nu_l} = \frac{1 - N\delta_{kl}}{N-1}.$$

On the import side,

$$\sum_{j \neq i} (dp_{ij} - d\log P_{M_i}) = 0,$$

so the $(1 - \rho)$ terms cancel when (47) is summed over $j \neq i$. Thus,

$$\sum_{j \neq i} d\log s_{ij} = (N-1)(1-\eta)\alpha_D d\log P_{M_i}.$$

Substituting into (49) gives

$$dTB_i = \frac{m}{N-1} \left[\sum_{k \neq i} (d\log s_{ki} + d\nu_k) - (N-1)(1-\eta)\alpha_D d\log P_{M_i} - (N-1) d\nu_i \right].$$

Fix $l \neq i$ and differentiate term by term:

$$\frac{\partial}{\partial \nu_l} \sum_{k \neq i} \log s_{ki} = -(1-\rho) \frac{N-2}{N-1} - \frac{(1-\eta)\alpha_D}{N-1}.$$

The export income terms contribute one, the import-side price-index term contributes $-(1-\eta)\alpha_D$, and the import income term contributes zero. Collecting terms gives

$$\tilde{J}_{il} = \frac{m}{(N-1)^2} \left[(N-1) - (1-\rho)(N-2) + N\alpha_D(\eta-1) \right] = \frac{m}{(N-1)^2} D_N.$$

The diagonal follows from (10):

$$\tilde{J}_{ii} = -(N-1)\tilde{J}_{il} = -\frac{m}{N-1} D_N.$$

At $N = 2$ this gives $\tilde{J}_{AB} = mD_2$, the coefficient in (25); deleting row and column A at $N = 3$ reproduces J in (56). $\square$

The two-country case. At the balanced free-trade baseline with normalized incomes, $s_{AB} = s_{BA} = m$. At $N = 2$ each country has a single import origin, so $dp_{AB} = d\log P_{MA} = d\nu_B + d\tau$ and $dp_{BA} = d\log P_{MB} = -d\nu_B$. Log-differentiating (18) and (19), with incomes in the common currency,

$$\begin{aligned} d\log s_{AB} &= (1 - \eta)\alpha_D(d\nu_B + d\tau), & d\log s_{BA} &= -(1 - \eta)\alpha_D d\nu_B, \\ d\log \tilde{I}_A &= m d\tau, & d\log \tilde{I}_B &= d\nu_B. \end{aligned}$$

Substituting into (20) and using that the tariff wedge lowers the border value of A 's imports by $d\tau$,

$$d\text{TB}_A = m \left[(1 + 2\alpha_D(\eta - 1))d\nu_B + (1 - m + \alpha_D(\eta - 1))d\tau \right],$$

which is (25), with $D_2 = 1 + 2\alpha_D(\eta - 1)$ and $\rho_1^* = 1 - m + \alpha_D(\eta - 1)$, and (26) follows. For the elasticity form, hold income fixed and let $M(p)$ denote import volume at consumer price $p = p_{AB}$. Since import value is $s_{AB}I_A$,

$$\varepsilon_M \equiv -\frac{\partial \log M}{\partial \log p} = 1 - \frac{\partial \log s_{AB}}{\partial \log p} = 1 - (1 - \eta)\alpha_D = 1 + \alpha_D(\eta - 1).$$

By symmetry, $\varepsilon_X = 1 + \alpha_D(\eta - 1)$. Differentiating $\text{TB}_A(e, \tau) = X(1/e) - eM((1 + \tau)e)$ at $\tau = 0$ and using balanced trade, $X = eM$, gives

$$\frac{\partial \text{TB}_A}{\partial e} = M(\varepsilon_X + \varepsilon_M - 1).$$

Hence $\varepsilon_X + \varepsilon_M - 1 = 2\alpha_D(\eta - 1) + 1 = D_2$, so the stability condition $D_2 > 0$ is exactly $\varepsilon_X + \varepsilon_M > 1$. Since $\rho_1^* > 0$ by (23), the slope $-\rho_1^*/D_2$ is negative whenever $D_2 > 0$.

The tariff disturbance and the four eigenvalues. At unchanged exchange rates, the effect on partner i 's trade balance follows from (49). Since only i 's exports to A are directly affected,

$$d\text{TB}_i|_{\nu \text{ fixed}} = f_{Ai}(d\log s_{Ai} + d\log \tilde{I}_A - t_i) = \frac{m}{N-1} \left[-\rho_1^* \bar{t} - \rho(t_i - \bar{t}) \right]. \quad (51)$$

Reading off the coefficients on $\bar{t}$ and $\tilde{t}_i$ gives the tariff-effect eigenvalues (29). Deleting row and column A from (50) gives the exchange-rate adjustment matrix

$$-J = c(NI - \mathbf{1}\mathbf{1}'), \quad c \equiv \frac{mD_N}{(N-1)^2}, \quad (52)$$

whose eigenvalues are $j_S = c$ on $\text{span}\{\mathbf{1}\}$ and $j_D = Nc$ on the zero-sum subspace, which is (30). Dividing (29) by (30) gives the coefficients λ_S and λ_D of Theorem 3.1.

The unilateral tariff. For $\mathbf{t} = d\tau \mathbf{e}_B$, $\bar{t} = d\tau/(N-1)$, $\tilde{t}_B = \frac{N-2}{N-1}d\tau$ and $\tilde{t}_C = -\frac{d\tau}{N-1}$. By Theorem 3.1,

$$d\nu_B = \frac{\lambda_S + (N-2)\lambda_D}{N-1} d\tau = -\frac{N\rho_1^* + (N-2)\rho}{ND_N} d\tau, \quad d\nu_C = \frac{\lambda_S - \lambda_D}{N-1} d\tau = \frac{\rho - N\rho_1^*}{ND_N} d\tau,$$

while $d \log e_{BC} = d\nu_C - d\nu_B = -\lambda_D d\tau$ and $d\nu_A^E = \lambda_S d\tau/(N-1) = -(\rho_1^*/D_N) d\tau$. Since $\rho_1^* > 0$ by (23) and $D_N > 0$ under MML, these give (34)–(36).

Derivation of (37). With $\rho = \eta = \sigma$,

$$N\rho_1^* = N[1 + \alpha_D(\sigma - 1) - \alpha_T(1 - \alpha_D)],$$

so

$$N\rho_1^* - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T).$$

For $\alpha_D \geq 1/N$ both terms are nonnegative, with both zero only at $\alpha_D = 1/N$, $\alpha_T = 1$; hence $\rho = \sigma \leq N\rho_1^*$. Since ρ_1^* does not involve ρ ,

$$N\rho_1^* - \eta = \eta(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T).$$

At $\alpha_D = 1/N$,

$$N\rho_1^* - \sigma = (N-1)(1 - \alpha_T), \quad D_N = (N-1)\sigma,$$

so by (35),

$$\frac{d \log e_{AC}}{d\tau} = -\frac{1 - \alpha_T}{N\sigma}.$$

Relative treatment across partners. By (28), $d\nu_j - d\nu_k = \lambda_D(t_j - t_k)$ with $\lambda_D < 0$, so $t_j > t_k$ implies $d\nu_j < d\nu_k$. With $\bar{t} > 0$, A depreciates against partner j if and only if

$$\frac{\rho}{N}(\bar{t} - t_j) > \rho_1^* \bar{t} \quad \Longleftrightarrow \quad \frac{t_j}{\bar{t}} < 1 - \frac{N\rho_1^*}{\rho}. \quad (53)$$

To see this, note that $d\nu_j > 0$ if and only if $\lambda_S \bar{t} + \lambda_D(t_j - \bar{t}) > 0$. Substituting λ_S and λ_D from Theorem 3.1 and dividing by $(N-1)/D_N > 0$ gives $-\rho_1^* \bar{t} - \frac{\rho}{N}(t_j - \bar{t}) > 0$, which rearranges to (53). The right-hand side of the second form is positive only if $\rho > N\rho_1^*$, so depreciation against a relatively lightly taxed partner requires both $\rho > N\rho_1^*$ and a sufficiently large tariff differential.

Tariffs on several partners. If A raises its tariff by $d\tau$ on a set of k partners, $1 \leq k \leq N-1$, then for every untargeted partner C and every targeted partner B ,

$$\frac{d \log e_{AC}}{d\tau} = \frac{k(\rho - N\rho_1^*)}{ND_N}, \quad \frac{d \log e_{AB}}{d\tau} = -\frac{k\rho_1^*}{D_N} - \frac{(N-1-k)\rho}{ND_N}, \quad \frac{d\nu_A^E}{d\tau} = -\frac{k\rho_1^*}{D_N}. \quad (54)$$

For every $k \leq N - 2$, the bystander response has the same threshold $\rho > N\rho_1^*$. When all partners face the same tariff,

$$\frac{d \log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} = \frac{d\nu_A^E}{d\tau} \quad \forall j. \quad (55)$$

All bilateral rates then move together, and substitution across foreign origins plays no role.

To derive these, let $t_j = d\tau$ on the k targets and zero otherwise, so that

$$\bar{t} = \frac{k d\tau}{N-1}, \quad \tilde{t}_B = \frac{N-1-k}{N-1} d\tau$$

for a target, and

$$\tilde{t}_C = -\frac{k d\tau}{N-1}$$

for a bystander. Substituting into (28) gives

$$d\nu_C = \frac{k d\tau}{N-1}(\lambda_S - \lambda_D) = \frac{k d\tau}{ND_N}(\rho - N\rho_1^*),$$

together with the target and NEER responses in (54). For every $k \leq N - 2$, the sign of the bystander response is therefore determined by $\rho - N\rho_1^*$. At $k = N - 1$ there is no bystander and $\tilde{t} = 0$, giving (55).

B.1 Three-country illustration

At $N = 3$, (52) and (51) give the unilateral-tariff system explicitly:

$$J = -\frac{mD_3}{4} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad F = -\frac{m}{4} \begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}, \quad D_3 = 3\alpha_D(\eta - 1) + \rho + 1. \quad (56)$$

With $D_3 > 0$, (34)–(35) read

$$\frac{d \log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D_3} < 0, \quad \boxed{\frac{d \log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D_3}}, \quad \frac{d \log e_{BC}}{d\tau} = \frac{2\rho}{3D_3} > 0. \quad (57)$$

Thus A appreciates against B , B depreciates against C , and A depreciates against C if and only if $\rho > 3\rho_1^*$. By (9), $d\nu_C/d\tau \propto F_B + 2F_C \propto \rho - 3\rho_1^*$, so the sign of C 's own disturbance F_C , which is positive once $\rho > \rho_1^*$, does not settle the sign of $d\nu_C$.

Figure 3 plots the two equilibrium conditions,

$$\text{TB}_B = 0 : \quad d\nu_C = 2 d\nu_B + \frac{\rho + \rho_1^*}{D_3} d\tau, \quad \text{TB}_C = 0 : \quad d\nu_C = \frac{1}{2} d\nu_B + \frac{\rho - \rho_1^*}{2D_3} d\tau. \quad (58)$$

In the intermediate case $\rho_1^* < \rho < 3\rho_1^*$, C gains from trade diversion at unchanged exchange rates and the $\text{TB}_C = 0$ locus shifts up. But the larger shift of $\text{TB}_B = 0$ drives $d\nu_B$ down, and the resulting movement along the upward-sloping $\text{TB}_C = 0$ locus more than offsets that shift: A still appreciates against C .

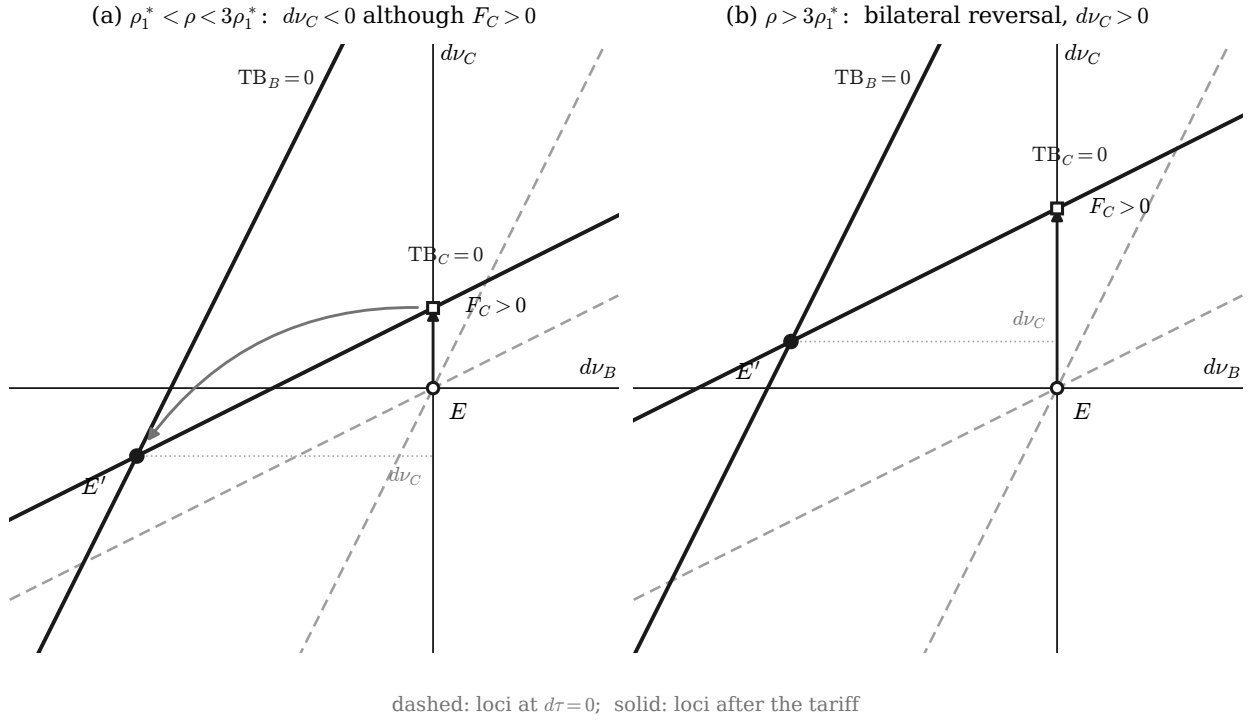


Figure 3: Three-country equilibrium adjustment. The loci are (58); dashed lines are the pre-tariff loci through the initial equilibrium E , solid lines the post-tariff loci intersecting at E' . The open square is the value of $d\nu_C$ that would clear C 's trade balance at $d\nu_B = 0$; the curved arrow in panel (a) is the subsequent movement along $TB_C = 0$. Panel (a): $\rho_1^* < \rho < 3\rho_1^*$. Panel (b): $\rho > 3\rho_1^*$. Drawn at $\alpha_D = 0.7$, $\alpha_T = 0.6$, $\eta = 1.5$, with $\rho = 2.2$ in panel (a) and $\rho = 5$ in panel (b), so that $\rho_1^* = 1.17$ and $3\rho_1^* = 3.51$.

C Real exchange rates

This appendix gives the real exchange-rate counterparts of the results in Sections 3.3–3.4. Real effective exchange-rate changes use the weights in (21), $dq_i^E \equiv \sum_{j \neq i} w_{ij} d \log q_{ij}$.

From (22) and (17),

$$d \log q_{Aj} = d\nu_j + \alpha_T (d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}}).$$

For a tariff vector $\mathbf{t}$, symmetry implies

$$d \log P_{M_A} = d\nu_A^E + \bar{t}, \quad d \log P_{M_j} = d\nu_A^E - \frac{N}{N-1} d\nu_j$$

for a partner that imposes no tariff. Hence

$$\begin{aligned} d \log q_{Aj} &= \left(1 - \frac{mN}{N-1}\right) d\nu_j - m \bar{t}, \\ dq_A^E &= \left[\left(1 - \frac{mN}{N-1}\right) \lambda_S - m\right] \bar{t}, \\ d \log q_{Aj} - dq_A^E &= \left(1 - \frac{mN}{N-1}\right) \lambda_D \tilde{t}_j. \end{aligned} \tag{59}$$

Thus average protection determines the real effective exchange-rate response, while relative treatment determines bilateral deviations around it, subject to the price-index adjustment in (59).

For a unilateral tariff, $\bar{t} = d\tau/(N-1)$ and

$$d \log q_{Aj} = \left(1 - \frac{mN}{N-1}\right) d\nu_j - \frac{m}{N-1} d\tau, \quad j = B, C. \tag{60}$$

Setting $d \log q_{AC} = 0$ and using (35), for $m < 1/N$ the real bilateral reversal threshold is

$$\rho_q^*(N) = \frac{[(N-1) - mN] N \rho_1^* + mN [N \alpha_D (\eta - 1) + 1]}{(N-1)(1 - mN)}. \tag{61}$$

Averaging (60) gives

$$\frac{dq_A^E}{d\tau} = -\left(1 - \frac{mN}{N-1}\right) \frac{\rho_1^*}{D_N} - \frac{m}{N-1} < 0 \quad \text{for all } \rho, N \text{ } (m < \frac{N-1}{N}). \tag{62}$$

Proposition C.1 (Real exchange rates). *At the symmetric benchmark of the nested-CES model, consider a unilateral tariff.*

- (i) *If $m < 1/N$ and MML holds at $\rho = \rho_q^*(N)$, the real bilateral reversal threshold exceeds the nominal threshold, $\rho_q^*(N) > N \rho_1^*$.*
- (ii) *If MML holds and $m < (N-1)/N$, the REER appreciates, $dq_A^E/d\tau < 0$.*

Both results reflect the direct price-index effect of the tariff. At a given nominal exchange rate, the tariff raises A 's consumer prices, making a real bilateral depreciation harder to obtain and

reinforcing the effective appreciation.

D Asymmetry

The disturbance-adjustment representation (6) and MML (8) do not require symmetry, and at any balanced baseline the linearized system (44)–(49) continues to hold with baseline shares and incomes as coefficients: b_{ij} , the share of origin j in i 's import bundle; h_i , the home share of i 's tradable expenditure; and $y_i \equiv \tilde{I}_i$, i 's income in the common currency. What requires symmetry is the collapse of the response to the two coefficients λ_S and λ_D of Proposition 2.3. Away from symmetry, G and $-J$ generally no longer act separately on average protection and relative treatment, so changes in relative treatment can affect the effective exchange rate. The unconditional NEER appreciation and the linear thresholds of Section 3 therefore need not survive. This appendix records what does.

Bilateral thresholds. The elasticity ρ enters the share responses (47) only through $(1 - \rho)$, so every entry of J and F is linear in ρ . Each entry of $\text{adj}(-J)$ is therefore a polynomial of degree at most $N - 2$ in ρ , and each numerator $[\text{adj}(-J)F]_i$ in (6) a polynomial of degree at most $N - 1$. Since $\det(-J) > 0$ under MML, a bilateral response can change sign only at a root of this numerator. Symmetry reduces the problem to the two coefficients of Proposition 2.3, yielding the linear thresholds (36) and (57). At a general asymmetric baseline the $N = 3$ numerator is already quadratic in ρ :

$$\text{sign} \frac{d \log e_{AC}}{d\tau} = \text{sign} (c_2 \rho^2 + c_1 \rho + c_0), \quad c_2 = b_{AB} b_{AC} b_{CA} b_{CB} (1 - h_A)(1 - h_C) y_A y_C. \quad (63)$$

The coefficients c_1 and c_0 depend on the full pattern of trade shares and country sizes. Provided MML continues to hold, $c_2 > 0$ whenever A and C each import from both foreign origins, so sufficiently strong substitution across import origins still reverses e_{AC} . When positive roots exist and MML continues to hold, the response is positive beyond the largest positive root, which equals $3\rho_1^*$ at the symmetric point. Away from symmetry there may be no positive root or several, so there is no universal bilateral threshold, and for $N \geq 4$ a response may change sign more than once as ρ varies.

E Correspondence across N

All entries are at the symmetric benchmark of the nested-CES model.

object	$N = 2$	$N = 3$	general N
adjustment condition	$D_2 = \varepsilon_X + \varepsilon_M - 1 > 0$	$D_3 > 0$	$D_N > 0$ (MML)
tariff-disturbance coefficients g_S, g_D (29)	$-m\rho_1^*, -$	$-m\rho_1^*/2, -m\rho/2$	$-m\rho_1^*/(N-1), -m\rho/(N-1)$
adjustment coefficients j_S, j_D (30)	$mD_2, -$	$mD_3/4, 3mD_3/4$	$mD_N/(N-1)^2, NmD_N/(N-1)^2$
disturbance and adjustment ratios $g_D/g_S, j_D/j_S$	$-$	$\rho/\rho_1^*, 3$	$\rho/\rho_1^*, N$
unilateral bystander threshold	none	$3\rho_1^*$	$N\rho_1^*$
unilateral NEER response	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$
trade-war threshold	$-$	ρ_1^*	ρ_1^*

F Proofs

Throughout, $m \equiv \alpha_T(1 - \alpha_D)$ and all derivatives are evaluated at the balanced free-trade baseline of Appendix B.

Proof of Lemma 2.2. Scaling every currency value e_{Al} by a common factor leaves all relative prices in (44) unchanged and scales every common-currency income, and hence every flow f_{ij} and trade balance TB_i , by the same factor. Thus TB_i is homogeneous of degree one in the currency values. By Euler's theorem, $\sum_l \tilde{J}_{il} = TB_i$, which is zero at the balanced baseline, so $\sum_l \tilde{J}_{il} = 0$ for all i . Equivalently, in (49), a common shift in all ν_l affects only the income terms, whose export and import contributions cancel by baseline balance. Summing (20) over i gives $\sum_i TB_i \equiv 0$, so $\sum_i \tilde{J}_{il} = 0$ for all l . Under Assumption 1, $\tilde{J}_{il} \geq 0$ for $l \neq i$, so $\tilde{J}_{ii} = -\sum_{l \neq i} \tilde{J}_{il} \leq 0$. Deleting row and column A then gives

$$|J_{ii}| - \sum_{l \neq i, A} J_{il} = \tilde{J}_{iA} \geq 0, \quad (64)$$

with strict inequality in every row for which $\tilde{J}_{iA} > 0$.

Thus $-J$ is a Z-matrix with nonpositive off-diagonal entries, nonnegative diagonal entries, weak diagonal dominance, irreducibility, and strict diagonal dominance in at least one row. Its diagonal is strictly positive, since $|J_{ii}| = 0$ would force row i of J to vanish, contradicting irreducibility. Write $-J = D - B$, where $D = \text{diag}(-J) > 0$ and $B \geq 0$. Diagonal dominance implies that every row sum of $D^{-1}B$ is at most one and that at least one is strictly below one. With irreducibility, its spectral radius therefore satisfies $r \equiv \rho(D^{-1}B) < 1$. Hence

$$(-J)^{-1} = (I - D^{-1}B)^{-1}D^{-1} = \sum_{k \geq 0} (D^{-1}B)^k D^{-1} \geq 0.$$

Nonsingularity follows from $r < 1$. Moreover,

$$\det(-J) = \det D \cdot \det(I - D^{-1}B) > 0.$$

□

Proof of Proposition 2.3. Let M be an $(N-1) \times (N-1)$ matrix with $PMP' = M$ for every permutation matrix P . Conjugating by the transposition of i and j gives $M_{ii} = M_{jj}$, and conjugating by a permutation that sends the ordered pair (i, j) to (k, l) , $i \neq j$, $k \neq l$, gives $M_{ij} = M_{kl}$. Hence

$$M = aI + b\mathbf{1}\mathbf{1}'.$$

Since $\mathbf{1}'\mathbf{1} = N-1$,

$$M\mathbf{1} = [a + (N-1)b]\mathbf{1},$$

while $Mv = av$ for every v satisfying $\mathbf{1}'v = 0$. This gives (16) for $-J$ and G .

For $j_S, j_D \neq 0$,

$$(-J)^{-1} = \frac{1}{j_D}I + \frac{1}{N-1}\left(\frac{1}{j_S} - \frac{1}{j_D}\right)\mathbf{1}\mathbf{1}',$$

as direct multiplication verifies. Thus $(-J)^{-1}$ and G share the constant and zero-sum eigenspaces, with response coefficients g_S/j_S and g_D/j_D . Writing

$$\mathbf{t} = \bar{t}\mathbf{1} + \tilde{t}, \quad \mathbf{1}'\tilde{t} = 0,$$

gives

$$d\boldsymbol{\nu} = \frac{g_S}{j_S}\bar{t}\mathbf{1} + \frac{g_D}{j_D}\tilde{t},$$

and averaging with equal weights yields

$$d\nu_A^E = \frac{g_S}{j_S}\bar{t}.$$

For (15), the off-diagonal entry of $-J$ is

$$b_J = \frac{j_S - j_D}{N-1},$$

so $-J$ has the Z-matrix sign pattern if and only if $j_D \geq j_S$. Given $j_D \geq j_S > 0$, the entries of $(-J)^{-1}$ above are nonnegative, so MML holds. Conversely, if $j_S \leq 0$, either $-J$ is singular or its inverse cannot be elementwise nonnegative. At $N = 2$ the zero-sum subspace is trivial and the condition reduces to $j_S > 0$.

Finally, for any two partners $j, k \neq A$,

$$d \log e_{jk} = d\nu_k - d\nu_j = \lambda_D(\tilde{t}_k - \tilde{t}_j) = \lambda_D(t_k - t_j),$$

which completes (14). □

Proof of Theorem 3.1. At the symmetric benchmark relabeling A 's partners leaves the reduced system unchanged, so Proposition 2.3 applies. Appendix B derives (50), (29) and (30). By (30), $j_S = mD_N/(N-1)^2$ and $j_D = Nj_S$, so $j_D \geq j_S$ always and $j_S > 0$ if and only if $D_N > 0$; by (15), MML is therefore equivalent to $D_N > 0$. If $\eta \geq 1$ then every term of (27) is nonnegative and the last is one, so $D_N > 0$. Given $D_N > 0$, dividing (29) by (30) gives the coefficients of Theorem 3.1,

$$\lambda_S = \frac{g_S}{j_S} = -\frac{m\rho_1^*}{N-1} \cdot \frac{(N-1)^2}{mD_N} = -\frac{(N-1)\rho_1^*}{D_N}, \quad \lambda_D = \frac{g_D}{j_D} = -\frac{(N-1)\rho}{ND_N},$$

both negative since $\rho_1^* > 0$ by (23) and $\rho > 0$. Equation (28) and the three statements following it are then (13) and (14). $\square$

Proof of Proposition 4.1(i). Let primes denote responses to the mirror shock $t_{BA} = d\tau$. Relabeling $A \leftrightarrow B$ in (34) gives

$$d \log e'_{BA} = -\frac{N\rho_1^* + (N-2)\rho}{ND_N}, \quad d \log e'_{BC} = \frac{\rho - N\rho_1^*}{ND_N}.$$

Since $e_{AB} = 1/e_{BA}$ and $e_{AC} = e_{AB}e_{BC}$,

$$d \log e'_{AB} = \frac{N\rho_1^* + (N-2)\rho}{ND_N}, \quad d \log e'_{AC} = \frac{(N-1)\rho}{ND_N}.$$

By linearity,

$$d \log e_{AB}|^w = 0,$$

and

$$d \log e_{AC}|^w = \frac{\rho - N\rho_1^*}{ND_N} + \frac{(N-1)\rho}{ND_N} = \frac{\rho - \rho_1^*}{D_N}.$$

In terms of Theorem 3.1, the three components are A 's average protection, A 's relative treatment, and B 's relative treatment:

$$d \log e_{AC}|^w = -\frac{\rho_1^*}{D_N} + \frac{\rho}{ND_N} + \frac{(N-1)\rho}{ND_N} = \frac{\rho - \rho_1^*}{D_N}.$$

By symmetry between A and B , the same expression applies to $d \log e_{BC}|^w$. Averaging gives

$$\left. \frac{d\nu_A^E}{d\tau} \right|^w = \frac{N-2}{N-1} \frac{\rho - \rho_1^*}{D_N}.$$

Under MML, $D_N > 0$, so the signs are determined by $\rho - \rho_1^*$. $\square$

Proof of Proposition 4.1(ii). Superpose the N tariff vectors. A 's tariff vector is uniform,

$$\bar{t}^{(A)} = d\tau, \quad \tilde{t}^{(A)} = 0,$$

and contributes $\lambda_S d\tau$ to every $d \log e_{Aj}$.

Partner j taxes only A . Its contribution to $d \log e_{Aj}$ is

$$-[\lambda_S + (N-2)\lambda_D] \frac{d\tau}{N-1},$$

while each of the $N-2$ other partners contributes $-\lambda_D d\tau$. Summing,

$$\begin{aligned} d \log e_{Aj} &= \lambda_S d\tau - [\lambda_S + (N-2)\lambda_D] \frac{d\tau}{N-1} - (N-2)\lambda_D d\tau \\ &= \frac{(N-2)(\rho - \rho_1^*)}{D_N} d\tau. \end{aligned}$$

The expression is the same for every j , so it also equals the NEER response, and all cross rates among A 's partners are unchanged. $\square$

Proof of Proposition C.1. From (22) and (17), only tradable prices move, so

$$d \log q_{Aj} = d\nu_j + \alpha_T (d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}}),$$

with

$$d \log P_{T_i}^{\text{own}} = (1 - \alpha_D) d \log P_{M_i}.$$

For a tariff vector $\mathbf{t}$,

$$d \log P_{M_A} = \frac{1}{N-1} \sum_j (d\nu_j + t_j) = d\nu_A^E + \bar{t},$$

while for a partner that imposes no tariff,

$$d \log P_{M_j} = d\nu_A^E - \frac{N}{N-1} d\nu_j.$$

Substitution gives (59), and $\bar{t} = d\tau/(N-1)$ gives (60).

For (i), setting $d \log q_{AC} = 0$ and substituting $d\nu_C$ from (35) gives

$$[(N-1) - mN] \frac{\rho - N\rho_1^*}{N} = mD_N(\rho).$$

For $m < 1/N$, solving for ρ gives (61). The coefficient $(N-1) - mN$ on the left is then positive, and MML at $\rho = \rho_q^*(N)$ gives $D_N > 0$ on the right, so $\rho_q^*(N) > N\rho_1^*$. For (ii), averaging (60) and using (33) gives (62); under MML $D_N > 0$, and for $m < (N-1)/N$ both terms are negative. $\square$